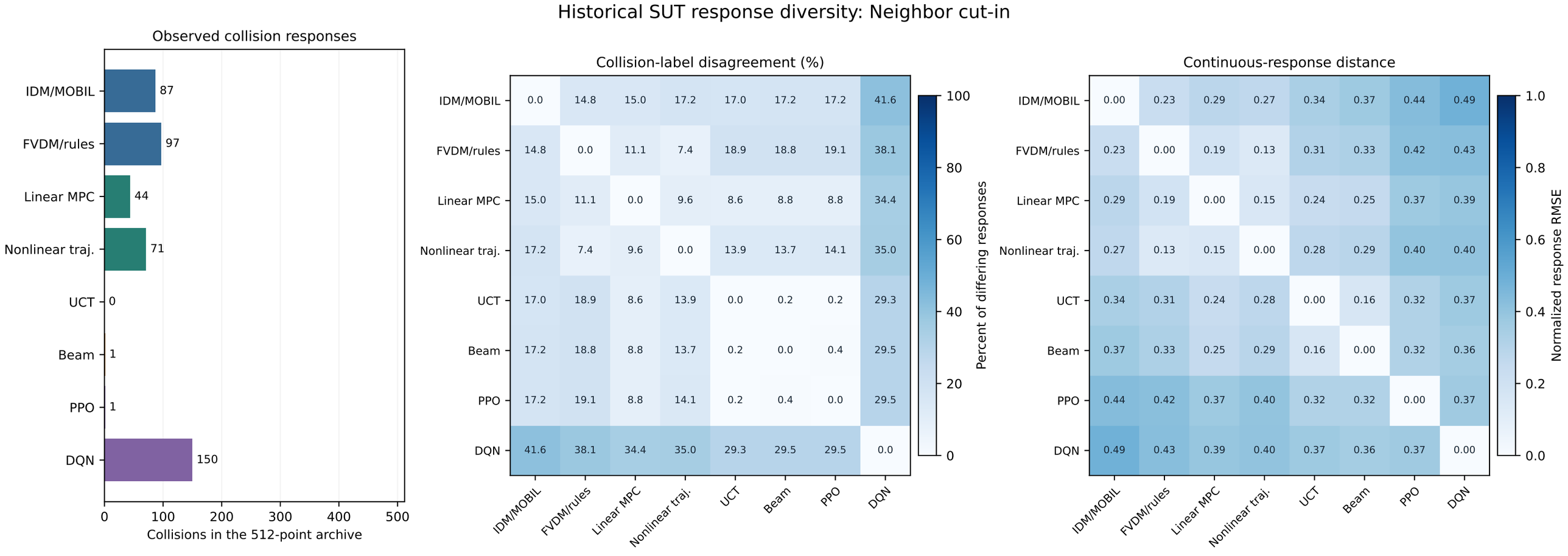
Historical SUT response diversity: Neighbor cut-in
Observed collision responses
IDM/MOBIL
FVDM/rules
Linear MPC
Nonlinear traj.
UCT
Beam
PPO
DQN
87
97
44
71
0
1
1
150
0
100
200
300
400
500
Collisions in the 512-point archive
Collision-label disagreement (%)
0.0 14.8 15.0 17.2 17.0 17.2 17.2 41.6
14.8 0.0 11.1 7.4 18.9 18.8 19.1 38.1
15.0 11.1 0.0 9.6 8.6 8.8 8.8 34.4
17.2 7.4 9.6 0.0 13.9 13.7 14.1 35.0
17.0 18.9 8.6 13.9 0.0 0.2 0.2 29.3
17.2 18.8 8.8 13.7 0.2 0.0 0.4 29.5
17.2 19.1 8.8 14.1 0.2 0.4 0.0 29.5
41.6 38.1 34.4 35.0 29.3 29.5 29.5 0.0
Percent of differing responses
100
80
60
40
20
0
Continuous-response distance
0.00 0.23 0.29 0.27 0.34 0.37 0.44 0.49
0.23 0.00 0.19 0.13 0.31 0.33 0.42 0.43
0.29 0.19 0.00 0.15 0.24 0.25 0.37 0.39
0.27 0.13 0.15 0.00 0.28 0.29 0.40 0.40
0.34 0.31 0.24 0.28 0.00 0.16 0.32 0.37
0.37 0.33 0.25 0.29 0.16 0.00 0.32 0.36
0.44 0.42 0.37 0.40 0.32 0.32 0.00 0.37
0.49 0.43 0.39 0.40 0.37 0.36 0.37 0.00
Normalized response RMSE
1.0
0.8
0.6
0.4
0.2
0.0